

Online Appendix

Proof of Lemma 1. Proof by induction. When $t = T$,

$$V_T^G(p, \emptyset) = pR(\emptyset, 1) + (1 - p)R(\emptyset, 0),$$

which is linear in p and thus convex in p . Next, fix any $t < T$, and assume that $V_{t+1}(p, \emptyset)$ is convex in p . We want to show that $V_t(p, \emptyset)$ is convex in p . This is equivalent to showing that for all $p, p' \in [0, 1]$ and $\lambda \in [0, 1]$,

$$\lambda V_t(p, \emptyset) + (1 - \lambda)V_t(p', \emptyset) \geq V_t(\bar{p}),$$

where $\bar{p} \equiv \lambda p + (1 - \lambda)p'$. To this end, fix a p, p' , and $\lambda \in [0, 1]$ and define the following binary signal $b \in \{0, 1\}$ on θ :

$$Pr(b = 1 | \theta = 1) = \frac{p\lambda}{\bar{p}}, \quad Pr(b = 1 | \theta = 0) = \frac{(1 - p)\lambda}{1 - \bar{p}},$$

Now define the following two signals σ and $\tilde{\sigma}$:

$$\sigma : \{0, 1\} \rightarrow \Delta([0, \infty]), \text{ where } \sigma(\theta) = F(\cdot | \theta)$$

$$\tilde{\sigma} : \{0, 1\} \rightarrow \Delta([0, \infty] \times \{0, 1\}), \text{ where } \tilde{\sigma}(\theta) = \tilde{F}(\cdot, \cdot | \theta),$$

and for $b^* \in \{0, 1\}$, $\tilde{F}(s, b^*) = F(s | \theta) Pr(b \leq b^* | \theta)$. Note that $\tilde{\sigma}$ Blackwell dominates σ . Now assuming that the agent has prior belief p let $G_t(\cdot | p)$ and $\tilde{G}_t(\cdot | p)$ denote the distribution of posteriors after observing σ and $\tilde{\sigma}$, respectively. It follows from the Law of Iterated Expectations that:

$$\tilde{G}_t(q | \bar{p}) = Pr(b = 1 | \bar{p})G_t(q | p) + Pr(b = 0 | \bar{p})G_t(q | p') = \lambda G_t(q | p) + (1 - \lambda)G_t(q | p'). \quad (18)$$

Now, note that since $V_{t+1}(p, \emptyset)$ is convex in p and $V_{t+1}(p, 1) = pR(t + 1, 1) + (1 - p)R(t +$

$1, 0)$ is linear in p , $V_{t+1}(p) = \max\{V_{t+1}(p, \emptyset), V_{t+1}(p, 1)\}$ is also convex in p . Thus

$$\begin{aligned}
V_t(\bar{p}, \emptyset) &= \int_0^1 V_{t+1}(q) dG_t(q|\bar{p}) \\
&\leq \int_0^1 V_{t+1}(q) d\tilde{G}_t(q|\bar{p}) \\
&= \lambda \int_0^1 V_{t+1}(q) dG_t(q|p) + (1 - \lambda) \int_0^1 V_{t+1}(q) dG_t(q|p') \\
&= \lambda V_t(p, \emptyset) + (1 - \lambda) V_t(p', \emptyset),
\end{aligned}$$

where the inequality follows from Blackwell's (1953) Theorem, and the second equality follows from (18). $\square$

Proof of Lemma 2. Fix a t , and assume by induction that (1) holds for all $s > t$.

First, consider the case where $A_s^B = 1$ for some $s < t$. By informativeness, $Pr(\tau < t | i = G) < 1$. Thus $R_{t-1} = 1$. It follows that $V_s^R(p, 1) = V_s^R(p, \emptyset) = 1$ for all $s \geq t$. Since

$$V_t^D(1, 1) - V_t^D(1, \emptyset) > 0 \text{ and } V_t^D(0, 1) - V_t^D(0, \emptyset) < 0,$$

(1) holds in t .

Next, suppose $A_s^B < 1$ for all $s < t$. Consider the case where $A_t^B = 1$. By the informativeness property, $Pr(\tau = t | \tau \geq t, i = G) < 1$. Thus,

$$V_t^R(p, 1) < V_t^R(p, \emptyset) = 1 \text{ for all } p, \quad (19)$$

and so $V_t(0, 1) < V_t(0, \emptyset)$. To show that (1) holds in t , it remains to show that $V_t(1, 1) > V_t(1, \emptyset)$. Assume by contradiction $V_t(1, 1) \leq V_t(1, \emptyset)$. If $t = T$, then $V_t(p_0, \emptyset) > V_t(p_0, 1)$, and thus $A_t^B = 0$. Contradiction. If $t < T$, it follows from the inductive assumption that $V_t(1, \emptyset) = V_{t+1}(1, 1)$. So,

$$V_{t+1}(1, 1) \leq V_t(1, 1).$$

By (19) and the fact that $V_{t+1}^D(0, 1) > V_t^D(0, 1)$ it follows that $V_{t+1}(0, 1) > V_t(0, 1)$, and thus $V_{t+1}(p_0, 1) > V_t(p_0, 1)$. So, B can profitably deviate by choosing $A_t^B = 0$ and $A_{t+1}^B = 1$. Contradiction.

Next, consider the case where $A_t^B = 0$. By the informativeness property, $Pr(\tau = t | \tau \geq t, i = G) > 0$. Thus, $V_t^R(p, \emptyset) < V_t^R(p, 1) = 1$ for all p , and so $V_t(1, 1) > V_t(1, \emptyset)$. To show (1) holds in t , it remains to show $V_t(0, 1) < V_t(0, \emptyset)$. Assume by contradiction $V_t(0, 1) \geq$

$V_t(0, \emptyset)$. Since $V_t(p, 1)$ is linear in p and by [Lemma 1](#) $V_t(p, \emptyset)$ is convex in p ,

$$V_t(p, 1) > V_t(p, \emptyset) \text{ for all } p > 0.$$

It follows that

$$V_t(p_0, 1) > V_t(p_0, \emptyset) \geq V_t^B(p_0, \emptyset).$$

Thus B can profitably deviate by playing $A_t^B = 1$. Contradiction.

Finally, consider the case where $A_t^B \in (0, 1)$. I claim that $A_{t+1}^B > 0$ whenever $t < T$. Suppose not, by contradiction. By the informativeness property, $Pr(\tau = t + 1 | i = G) > 0$. First, consider the case where $V_1^D(p_0, 1) \leq 0$. Because $A_t^B \in (0, 1)$, $V_t^R(p_0, 1) < 1$ and because $A_{t+1}^B = 0$, $V_{t+1}^R(p_0, 1) = 1$. Thus, $V_t(p_0, 1) < V_{t+1}(p_0, 1)$. It follows that $A_t^B = 0$ and $A_{t+1}^B = 1$ is a profitable deviation.

Next, consider the case where $V_1^D(p_0, 1) > 0$. Since $V_{t+1}^R(p_0, 1) = 1$, $V_{t+1}(p_0, 1) > V_\tau(p_0, 1)$ for all $\tau \neq t + 1$, and thus $A_{t+1}^B = 1$ is a profitable deviation. It follows from the above statement that B is indifferent between acting at t and $t + 1$ when $t < T$. Thus,

$$V_t(p_0, 1) = V_{t+1}(p_0, 1) \tag{20}$$

Now, we want to show $V_t(1, 1) > V_t(1, \emptyset)$. Suppose not by contradiction. By the inductive assumption, $V_t(1, \emptyset) = V_{t+1}(1, 1)$ and $V_t(1, 1) \leq V_{t+1}(1, 1)$. Thus, by (20), $V_t(0, 1) \geq V_{t+1}(0, 1)$. If $X < 1$ this implies

$$R(t, 1) < R(t + 1, 1) \text{ and } R(t, 0) > R(t + 1, 0),$$

which is a violation of [Assumption 2](#). Now suppose $X = 1$. First, suppose $R(t, 1) < R(t + 1, 1)$. In order for (20) to be satisfied, $R(t, 0) > R(t + 1, 0)$, which again implies a violation of [Assumption 2](#). We can analogously show that we cannot have $R(t, 0) > R(t + 1, 0)$. So, it follows that $R(t, 1) = R(t + 1, 1)$ and $R(t, 0) = R(t + 1, 0)$. When $t = T$, this implies a failure of informativeness. When $t < T$, by the inductive assumption,

$$V_t(p, \emptyset) > V_{t+1}(p, 1) = V_t(p, 1) \text{ for all } p > 0.$$

Thus, $Pr(\tau = t | i = G, \theta) = 0$ for all θ . Since $A_t^B > 0$, this implies $R(t, \theta) = 0$ for all θ , and thus $V_t(p, 1) = 0$ for all p . But by informativeness, $R(s, \theta) > 0$ for some $s > t$, θ . Thus, B can profitably deviate by playing $A_t^B = 0$. $\square$

Proof of Corollary 1. Fix any $t < T$. First note

$$V_t(1, \emptyset) = V_{t+1}(1, 1) = R(t+1, 1),$$

where the first equality follows from Lemma 2. Next, note $V_t(1, 1) = R(t, 1)$. Finally, by Lemma 2, $V_t(1, 1) > V_t(1, \emptyset)$. Thus, $R(t, 1) > R(t+1, 1)$. The bad agent's indifference condition then implies $R(t, 0) < R(t+1, 0)$. □

Proof of Lemma 3. Because G plays a cutoff strategy, it suffices to show that $V_t^G(\hat{p}_t, 1) > V_t^G(\hat{p}_t, \emptyset)$. Note that

$$V_t^{D,G}(\hat{p}_t, 1) = \hat{V}_t^G(\hat{p}_t, 1) = \hat{V}_t^G(\hat{p}_t, \emptyset) \geq V_t^{D,G}(\hat{p}_t, \emptyset),$$

where the second equality follows from the definition of $\hat{p}_t$. Thus,

$$V_t^G(\hat{p}_t, 1) = (1-X)V_t^{D,G}(\hat{p}_t, 1) + XV_t^R(\hat{p}_t, 1) > (1-X)V_t^{D,G}(\hat{p}_t, \emptyset) + XV_t^R(\hat{p}_t, \emptyset) = V_t^G(\hat{p}_t, \emptyset),$$

where the strict inequality follows from the assumptions that $V_t^R(\hat{p}_t, 1) > V_t^R(\hat{p}_t, \emptyset)$ and $X > 0$. □

Proof of Lemma 4. Fix a t . Suppose by induction $p_s^* < 1$ for all $s < t$. We want to show $p_t^* < 1$. First consider the case where $A_s^B = 1$ for some $s < t$. This implies $X < 1$, otherwise, $A_s^B = 1$ cannot be part of a fixed point. It also implies $R_{t-1} = 1$, and thus $p_t^* = \hat{p}_t \in (0, 1)$. Next, consider the case where $A_s^B < 1$ for all $s < t$. First, suppose that $A_t^B = 0$. Then $R(t, \theta) = 1$ for all θ , so $p_t^* \leq \hat{p}_t < 1$. Next, suppose that $A_t^B > 0$. This implies that $V_t(1, 1) > V_t(1, \emptyset)$. Thus, $p_t^* < 1$. □

Proof of Lemma 5. Fix any fixed point x of Φ . First, I show that (x, R^x) satisfies Assumption 1 and Assumption 2. That Assumption 2 is satisfied follows from the assumption that G plays cutoff strategies in every t . To see why Assumption 1 holds, note by Lemma 4 that for all t , $p_t^* < 1$. Now fix a t and suppose that $A_s^B < 1$ for all $s < t$.

If $A_t^B = 0$, then

$$Pr(\tau = t | i = B, \theta, \tau \geq t) = 0 \text{ and } Pr(\tau = t | i = G, \theta, \tau \geq t) > 0 \text{ for all } \theta,$$

so [Assumption 1](#) is satisfied. If $A_t^B > 0$, then

$$Pr(\tau = t | i = B, \theta = 1, \tau \geq t) = Pr(\tau = t | i = B, \theta = 0, \tau \geq t).$$

However, since G plays a cutoff strategy,

$$Pr(\tau = t | i = G, \theta = 1, \tau \geq t) > Pr(\tau = t | i = G, \theta = 0, \tau \geq t),$$

so [Assumption 1](#) is again satisfied.

It remains to show that (x, R^x) is an equilibrium. It follows by definition that R^x is consistent with Bayes' Rule, given x . Next, I will show that given R^x , $(p_t^*)_{t=1}^T$ and $(A_t^B)_{t=1}^T$ are optimal for G and B , respectively. Since x is a fixed point, $A_t^B \in \Phi_t^B(x)$ for all t and the optimality of A_t^B follows from the definition of Φ_t^B . Next, consider G . First, we want to show that at any t , it is optimal for G to play a cutoff strategy. Now, note that because A_t^B is optimal under this fixed point for any t , the reasoning of [Lemma 2](#) applies here. Thus, (1) holds here. Because $V_t(p, \emptyset)$ is convex in p ([Lemma 1](#)) and $V_t(p, 1)$ is linear in p , playing cutoff strategy $\hat{p}_t$ is optimal. It remains to show that for all t , $\hat{p}_t = p_t^*$. Fix a t and suppose by induction that $\hat{p}_s = p_s^*$ for all $s > t$. By the definition of Φ_t^G , it follows that $\Phi_t^G(x) = p_t^* = \hat{p}_t$.

□

Proof of [Lemma 6](#). Fix a t , and assume A^G and A^B satisfy the given assumptions. We want to show that $R_t \in (0, 1)$. Proof by induction.

Base case: $s = 0$. $R_s = R_0 \in (0, 1)$ by assumption.

Induction step: For any $s \in \{1, \dots, t\}$, assume $R_{s-1} \in (0, 1)$. We want to show that $R_s \in (0, 1)$. It follows from Bayes' Rule that

$$R_s = \frac{1}{1 + \frac{1-R_{s-1}}{R_{s-1}} \frac{Pr(\tau \neq s | \tau \geq s, i=B)}{Pr(\tau \neq s | \tau \geq s, i=G)}}. \quad (21)$$

To show that $R_s \in (0, 1)$, it suffices to show that both the conditional probabilities in (21) lie in $(0, 1)$. In equilibrium,

$$Pr(\tau \neq s | \tau \geq s, i = B) = 1 - A_t^B \in (0, 1),$$

where $A_t^B \in (0, 1)$ holds by assumption. It remains to show that $Pr(\tau \neq s | \tau \geq s, i = G) \in$

$(0, 1)$. To this end, because the good agent is playing a cutoff strategy,

$$H_t(p_t|p_{t-1}) = \begin{cases} 0 & \text{for all } p < p_t^* \\ \frac{G_t(p_t|p_{t-1}) - G_t(p_t^*|p_{t-1})}{G_t(p_t^*|p_{t-1})} & \text{for all } p > p_t^* \end{cases}$$

We can write

$$Pr(\tau \neq s | \tau \geq s, i = G) = \int_0^1 G_t(p_t^*|p_{t-1}) dH_{t-1}(p_{t-1}). \quad (22)$$

Now, we make two observations:

1. $G_t(p_t^*|p_{t-1}) \in (0, 1)$ for all $p_{t-1} \in (0, 1)$.
2. $H_{t-1}(p_{t-1})$ is continuous in p_{t-1} , following from the continuity of $G_{t-1}(p_{t-1}|p_{t-2})$ in p_{t-1} .

It follows from the above two observations, combined with (22) that $Pr(\tau \neq s | \tau \geq s, i = G) \in (0, 1)$.

□

Proof of Lemma 7. Fix any $p \in (0, 1)$. It suffices to show that for each t , there exists $\bar{p}_t \in (0, 1)$ such that if $p_0 < \bar{p}_t$, then $p_t^* < p$ for any equilibrium under p_0 .

Suppose not, by contradiction. Let s be the last period such that the statement fails. Since by Proposition 3 $p_T^* = p_0$ in any equilibrium, $s < T$. Thus, there exists a $q > 0$ and a sequence of priors $(p_0^n)_{n=1}^\infty$ such that $\lim_{n \rightarrow \infty} p_0^n = 0$, where

$$p_s^{*,n} > q \text{ for all } n \text{ and } \lim_{n \rightarrow \infty} p_{s+1}^{*,n} = 0, \quad (23)$$

where $p_t^{*,n}$ is some equilibrium under p_0^n . Let superscript n refer to objects under this equilibrium.

It follows from B 's indifference (Proposition 3) that

$$\lim_{n \rightarrow \infty} [R^n(s+1, \theta=0) - R^n(s, \theta=0)] = 0. \quad (24)$$

Thus, in order for the $p_s^{*,n}$ to be optimal for all n

$$\lim_{n \rightarrow \infty} [R^n(s+1, \theta=1) - R^n(s, \theta=1)] = 0. \quad (25)$$

Now, it follows from Bayes' Rule that for any t ,

$$R^n(t, \theta) = \frac{1}{1 + \frac{1 - R_{t-1}^n}{R_{t-1}^n} X^n(t, \theta)},$$

where

$$X^n(t, \theta) \equiv \frac{A_t^{B,n} p_0^n}{Pr(\tau = t | \tau \not\leq t, i = G) Pr(\theta | \tau = t, i = G)}.$$

Thus, in order for (24) and (25) to hold,

$$\lim_{n \rightarrow \infty} X^n(s, \theta) = \lim_{n \rightarrow \infty} X^n(s + 1, \theta) \text{ for } \theta \in \{0, 1\}. \quad (26)$$

However, it follows from (23) that

$$\lim_{n \rightarrow \infty} \frac{Pr^n(\theta = 1 | \tau = s, i = G)}{Pr^n(\theta = 1 | \tau = s + 1, i = G)} = \infty \text{ while } \lim_{n \rightarrow \infty} \frac{Pr^n(\theta = 0 | \tau = s, i = G)}{Pr^n(\theta = 0 | \tau = s + 1, i = G)} = 0,$$

which implies that (26) fails. Contradiction.

□

Proof of Lemma 8. I begin by showing $A_t^B < 1$ for any t . To this end, I will prove the stronger statement that there exists $\underline{X} \in (0, 1)$ and $\underline{b} \in (0, 1)$ such that $A_t^B < \underline{b}$ in any equilibrium under any β when $X > \underline{X}$.

Proof by induction. Fix any t . Suppose there exists an $\underline{X}_{t-1}$ and $\underline{b}_{t-1} \in (0, 1)$ such that $A_s^B < \underline{b}_{t-1}$ for all $s < t$ when $X > \underline{X}_{t-1}$. This holds vacuously when $t = 1$. We want to show that there exists $\underline{X}_t \in (\underline{X}_{t-1}, 1)$ and $\underline{b}_t$ such that $A_s^B < \underline{b}_t$ for all $s \leq t$ when $X > \underline{X}_t$. Suppose not, by contradiction. Then, there exists a sequence of X , $(X_n)_{n=1}^\infty$ with $X_n \rightarrow 1$ where $A_t^{B,n} \rightarrow 1$ for some equilibrium strategy $A_t^{B,n}$ for B under some value of β under X_n . Let superscript n denote equilibrium objects under each such equilibrium. I claim $V_t^{B,n}$ must satisfy the following two properties:

1. There exists an N and $\underline{V} < 1$ such that $V_t^{B,n}(p_0, 1) < \underline{V}$ for all $n > N$.
2. $\lim_{n \rightarrow \infty} V_t^{B,n}(p_0, \emptyset) = 1$.

These two properties imply that there exists an N such that for all $n > N$, $V_t^{B,n}(p_0, 1) < V_t^{B,n}(p_0, \emptyset)$, thus implying that $A_t^{B,n} = 0$, a contradiction.

I begin by establishing 1. Since $X_n \rightarrow 1$, it suffices to show that there exists N and $\bar{R} < 1$ such that $R^n(t, 1) < \bar{R}$ for $n > N$. The inductive assumption implies there exists $\hat{R} < 1$ such that $R_{t-1}^n < \hat{R}$. Since $A_t^n \rightarrow 1$, there exists an N and $\underline{A} > 0$ such that for all

$n > N$, $A_t^{B,n} > \underline{A}$. Recall that

$$R^n(t, 1) = \frac{1}{1 + \frac{1-R_{t-1}}{R_{t-1}} \frac{p_0 A_t^{B,n}}{Pr^n(\tau=t|i=G, \tau \geq t, \theta=1)}}.$$

It thus follows that for all $n > N$,

$$R^n(t, 1) < \bar{R} \equiv \frac{1}{1 + \frac{1-\hat{R}}{\hat{R}} \underline{A} p_0} < 1.$$

Next, I establish 2. It suffices to show $R_t^n \rightarrow 1$. We know by assumption that $A_t^{B,n} \rightarrow 1$. Note that for any n such that $A_t^{B,n} = 1$, by the informativeness assumption, $p_t^{*,n} < 1$, and thus $R_t^n = 1$. Let m index the subsequence of $(X_n)_{n=1}^\infty$ such that $A_t^{B,n} < 1$. Note that if X_m is finite, then the statement holds trivially. Now suppose X_m is an infinite subsequence. $A_t^{B,m} < 1$ implies $V_t^{B,m}(p_0, 1) \leq V_t^{B,m}(p_0, \emptyset)$. This implies $V_t^{G,m}(p_0, 1) \leq V_t^{G,m}(p_0, \emptyset)$ and thus $p_t^{*,m} \geq p_0 > 0$. This then implies that there exists $q > 0$ such that $Pr^m(\tau \neq t | \tau \geq t, G) > q$. Furthermore, by the inductive assumption, $p_{s,m}^* > p_0$ for all m and $s < t$. So, there exists a $\hat{R} > 0$ such that $R_{t-1}^m > \hat{R}$ for all m . Thus,

$$\lim_{m \rightarrow \infty} R_t^m = \lim_{m \rightarrow \infty} \frac{1}{1 + \frac{1-R_{t-1}^m}{R_{t-1}^m} \frac{1-A_t^{B,m}}{Pr^m(\tau \neq t | \tau \geq t, i=G)}} = 1.$$

Now, I show that $A_t^B > 0$ for any t . Suppose not, by contradiction. Then there exists a sequence of X $(X_n)_{n=1}^\infty$ with $X_n \rightarrow 1$ where $A_t^{B,n} = 0$ for all n under some equilibrium strategy $A_t^{B,n}$ under X_n and some β . Since G acts with positive probability in equilibrium under any t by [Lemma 2](#), $R^n(t, \theta) = 1$ for all θ . Since $X_n \rightarrow 1$, $\lim_{n \rightarrow \infty} V_t^{B,n}(p_0, 1) = 1$. Thus, $\lim_{n \rightarrow \infty} V_t^{B,R,n}(p_0) = 1$. Since $R_0 < 1$, this implies that for n sufficiently large, B 's expected reputation at the end of the game exceeds R_0 . Thus R^n cannot be consistent with Bayes' Rule. Contradiction. $\square$

Proof of [Lemma 9](#). I begin by establishing the bound $\underline{R}$. By [Lemma 8](#), in every equilibrium under any β and t , $V_t^B(p_0) = V_t^B(p_0, \emptyset)$. This implies $V_t^G(p_0, 1) \leq V_t^G(p_0, \emptyset)$, and thus $p_t^* \geq p_0$ for all t . Thus there exists $\underline{p} \in (0, 1)$ such that $Pr(\tau = t | \theta = 0, i = G) \geq \underline{p}$ in any equilibrium, under any β . Namely, one such $\underline{p}$ is the probability $\tau = t$ given $\theta = 0$, $i = G$ and the good agent plays cutoff strategies $p_s^* = p_0$ for all $s < t$ and $p_t^* = \bar{p}_t$. Thus, in any equilibrium

$$R(t, 0) \geq \underline{R} \equiv \frac{1}{1 + \frac{1-R_0}{R_0} \frac{1}{\underline{p}}} > 0.$$

Now, I establish the bound $\bar{R}$. Suppose by contradiction that there does not exist such a $\bar{R}$. Then, there exists a sequence of equilibria, indexed by n (under possibly different values of β) such that

$$\lim_{n \rightarrow \infty} R^n(t, 1) = 1. \quad (27)$$

Now, recall that

$$R^n(t, 1) = \frac{1}{1 + \frac{1-R_0}{R_0} \frac{Pr^n(\tau=t|i=B, \theta=1)}{Pr^n(\tau=t|i=G, \theta=1)}}.$$

Thus, in order for (27) to hold, $\lim_{n \rightarrow \infty} Pr^n(\tau = t|i = B, \theta = 1) = 0$. Since $Pr^n(\tau = t|i = B, \theta = 0) = \frac{1-p_0}{p_0} Pr^n(\tau = t|i = B, \theta = 1)$,

$$\lim_{n \rightarrow \infty} Pr^n(\tau = t|i = B, \theta = 0) = 0.$$

However, as previously shown for all n , there exists $\underline{p} > 0$ such that

$$Pr^n(\tau = t|\theta = 0, i = G) \geq \underline{p}.$$

This implies

$$\lim_{n \rightarrow \infty} \frac{Pr^n(\tau = t|\theta = 0, i = B)}{Pr^n(\tau = t|\theta = 0, i = G)} = 0,$$

and therefore $\lim_{n \rightarrow \infty} R^n(t, 0) = 1$. Since $\lim_{n \rightarrow \infty} R^n(t, \theta) = 1$ for all θ , it follows that for n sufficiently large, B can profitably deviate by playing the pure strategy $A_t^{B,n} = 1$. Contradiction. $\square$

Proof of Proposition 5. First, we want to show that both players play cutoff strategies in every t . Note that the proofs of Lemma 1 and Lemma 2 do not rely on the assumption that B is unable to learn, and thus apply in this setting as well. Now note that Lemma 1 applies to B 's value function as well, i.e., $V_t^B(p, \emptyset)$ is convex in p . Next, because $V_t^G(\theta, a) = V_t^B(\theta, a)$ for all $\theta \in \{0, 1\}$ and $a \in \{\emptyset, 1\}$, Lemma 2 applies to B 's value function as well, i.e.,

$$V_t^B(1, 1) > V_t^B(1, \emptyset) \text{ and } V_t^B(0, \emptyset) > V_t^B(0, 1) \text{ for all } t.$$

That $p_T^{*,B} = p_T^{*,G}$ follows immediately from the fact that $V_T^B(p, a) = V_T^G(p, a)$ for all p, a . To show $p_t^{*,B} < p_t^{*,G}$ for all $t < T$, note $V_t^B(p, 1) = V_t^G(p, 1)$ for all t , but because G 's signal Blackwell dominates B 's signal and has a strictly larger support on the likelihood ratios:

$$V_t^B(p, \emptyset) < V_t^G(p, \emptyset) \text{ for all } t < T \text{ and } p \in (0, 1).$$

The statement follows immediately. $\square$

Proof of Proposition 6. We wish to extend the proof of Theorem 1 to the setting where the bad agent observes signals drawn from f^B . Thus, fix an f^B with support $(\underline{z}, \bar{z})$, where $\underline{z} > 0$ and $\bar{z} < \infty$. First, note that Lemma 3 extends immediately. Now, rather than letting A_t^B refer to the bad agent's strategy in t , given B 's strategy $(p_t^{*,B})_{t=1}^T$, let

$$A_t^B \equiv \Pr(\tau = t | \tau \geq t, i = B).$$

Let us begin by extending the proof of existence of $\bar{K}_1$ (again, the existence of $\bar{K}_0$ follows analogously). Following the proof of Theorem 1, there exists N and $\bar{W} < 0$ such that for all $n > N$, $W_s^{D,B,n}(\bar{p}_s) < \bar{W}$, where $\bar{p}_s < 1$ is the upper bound on the support of beliefs for the bad agent in s . Since $A_s^{B,n} \in (0, 1)$,

$$W_s^{R,B,n}(\bar{p}_s) > -\bar{W} > 0.$$

So it again follows that for all $n > N$,

$$R^n(s, 1) - R^n(s+1, 1) > -\bar{W},$$

a contradiction.

Next, we will extend the proof of the existence of $\bar{p}$. We will extend each step of the proof in turn. Again, we assume by contradiction that there exists a sequence $\{p_{0,n}\}_{n=1}^\infty$ where $p_{0,n} \in (0, 1)$ and $\lim_{n \rightarrow \infty} p_{0,n} = 0$, such that for all n $p_{t,n}^* \geq \hat{p}_t$. Hereafter, let $\bar{p}_{t,n}$ denote the upper bound on the support of beliefs that the bad agent might hold at t under prior $p_{0,n}$.

To extend step 1, we will instead show that there exists an N such that for all $n > N$, $A_{s,n}^B > 0$ for all $s \geq t$. The proof that $A_{t,n}^B > 0$ for all n follows identically. Next, consider $s > t$. First note $\lim_{n \rightarrow \infty} \bar{p}_{t,n} = 0$. Thus there exists N such that for all $n > N$,

$$\bar{p}_{t,n} K_1 + (1 - \bar{p}_{t,n}) K_0 < 0.$$

Now, suppose by contradiction that for some $n > N$, $A_{s,n}^B = 0$. Then we have that

$$V_{s,n}^B(\bar{p}_{t,n}, 1) > V_{t,n}^B(\bar{p}_{t,n}, 1),$$

which contradicts that $A_{t,n}^B > 0$. Hereafter, I redefine $\{p_{0,n}\}_{n=1}^\infty$ to be the sequence $\{p_{0,n}\}_{n=N+1}^\infty$.

To extend step 2, we note by step 1 that $A_{s,n}^B \in (0, 1)$ for all $s \geq t$ (that $A_{s,n}^B < 1$ is

immediate). Thus for all $s \geq t$ and n , there exists $p_{s,n} \in (\underline{p}_{T,n}, \bar{p}_{T,n})$ such that

$$V_{s,n}^B(p_{s,n}, 1) = V_{s,n}^B(p_{s,n}, \emptyset).$$

Since $\lim_{n \rightarrow \infty} \underline{p}_{T,n} = \lim_{n \rightarrow \infty} \bar{p}_{T,n} = 0$,

$$\lim_{n \rightarrow \infty} V_{s,n}^B(p_{s,n}, 1) - V_{s+1,n}^B(p_{s,n}, 1) = (1 - X)(\beta^s - \beta^{s+1})K_0 + X \lim_{n \rightarrow \infty} [R^n(s, 0) - R^n(s+1, 0)] = 0.$$

Thus step 2 follows from the same reasoning as in [Theorem 1](#).

To extend step 3, first define

$$Q_{n,t}^B \equiv \frac{\int_0^1 \int_{p_{t,n}^{*,B}}^1 (1 - p_t) dG_{t,n}^B(p_t | p_{t-1}) dH_{t-1,n}^B(p_{t-1})}{\int_0^1 \int_{p_{t,n}^{*,B}}^1 p_t dG_{t,n}^B(p_t | p_{t-1}) dH_{t-1,n}^B(p_{t-1})},$$

where $G_{t,n}^B$ and $H_{t,n}^B$ are defined analogously for the bad agent as they are for the good agent. We want to show

$$\lim_{n \rightarrow \infty} \frac{Q_{n,t}}{Q_{n,t}^B} = 0.$$

As before, $Q_{n,t} < \frac{1}{p_{t,n}^*} - 1$. Meanwhile, since $p_{t,n}^{*,B} < \bar{p}_{t,n}$ for all n ,

$$Q_{n,t}^B > \frac{\int_0^1 \int_{p_{t,n}^{*,B}}^1 dG_{t,n}^B(p_t | p_{t-1}) dH_{t-1,n}^B(p_{t-1})}{\int_0^1 \int_{p_{t,n}^{*,B}}^1 \bar{p}_{t,n} dG_{t,n}^B(p_t | p_{t-1}) dH_{t-1,n}^B(p_{t-1})} - 1 = \frac{1}{\bar{p}_{t,n}} - 1.$$

Recall that $\lim_{n \rightarrow \infty} \bar{p}_{t,n} = 0$. The statement follows immediately. By Step 2,

$$\frac{\int_0^1 \int_{p_{t,n}^{*,B}}^1 (1 - p_t) dG_{t,n}^B(p_t | p_{t-1}) dH_{t-1,n}^B(p_{t-1})}{\int_0^1 \int_{p_{t,n}^*}^1 (1 - p_t) dG_{t,n}(p_t | p_{t-1}) dH_{t-1,n}(p_{t-1})} \text{ is bounded.}$$

Thus,

$$\lim_{n \rightarrow \infty} \frac{\int_0^1 \int_{p_{t,n}^{*,B}}^1 p_t dG_{t,n}^B(p_t | p_{t-1}) dH_{t-1,n}^B(p_{t-1})}{\int_0^1 \int_{p_{t,n}^*}^1 p_t dG_{t,n}(p_t | p_{t-1}) dH_{t-1,n}(p_{t-1})} = \lim_{n \rightarrow \infty} \frac{\int_0^1 \int_{p_{t,n}^{*,B}}^1 (1 - p_t) dG_{t,n}^B(p_t | p_{t-1}) dH_{t-1,n}^B(p_{t-1})}{\int_0^1 \int_{p_{t,n}^*}^1 (1 - p_t) dG_{t,n}(p_t | p_{t-1}) dH_{t-1,n}(p_{t-1})} \frac{Q_{n,t}}{Q_{n,t}^B} = 0.$$

So, $\lim_{n \rightarrow \infty} R_n(t, 1) = 1$.

The extension of step 4 is immediate. A contradiction follows as in the proof of [Theorem 1](#).

□

Proof of Proposition 7. First, I argue Lemma 2 must hold when $T = \infty$ and $X = 1$. I now show $V_t(1, 1) > V_t(1, \emptyset)$ (that $V_t(0, 1) < V_t(0, \emptyset)$ follows analogously). Suppose by contradiction $V_t(1, 1) \leq V_t(1, \emptyset)$. This implies $R(t, 1) \leq R(\tau, 1)$ where $\tau \equiv \arg \max_{s > t, \emptyset} R(s, \theta = 1)$. If $R(t, 0) \leq R(\tau, 0)$ this implies that either G or B could profitably deviate (in G 's case, by never acting at t). If $R(t, 0) > R(\tau, 0)$, this implies that either G or B could profitably deviate, or in the case where $R(t, 1) < R(\tau, 1)$, there is a failure of Assumption 2.

That Lemma 2 holds implies that $R(t, \theta = 1)$ ($R(t, \theta = 0)$) is strictly decreasing (increasing) in t . Because they are also bounded, by the Monotone Convergence Theorem, there exists $R^1, R^0 \in [0, 1]$ such that

$$\lim_{t \rightarrow \infty} R(t, 1) = R^1 \text{ and } \lim_{t \rightarrow \infty} R(t, 0) = R^0. \quad (28)$$

This implies that there must exist a t such that $V_t(p, \emptyset)$ is convex, and thus the proof of Lemma 1 must extend to this setting. It follows that G plays a cutoff strategy $p_t^* \in (0, 1)$ and B mixes in every t . Now note that because of this

$$R(\emptyset, 1) < R(\emptyset, 0) \text{ and } R(t, 1) > R(t, 0) \text{ for all } t.$$

Thus, by (28), $\lim_{t \rightarrow \infty} p_t^* = 1$. In order for R to be consistent with Bayes' Rule, this implies that one of the two limits must hold:

$$\lim_{t \rightarrow \infty} R(t, 1) = 1 \quad \lim_{t \rightarrow \infty} R(t, 0) = 0.$$

However, this contradicts the fact that $R(t, 1)$ is strictly decreasing, and $R(t, 0)$ is strictly increasing in t . □

Proof of Claim 1. That $A^B \in (0, 1)$ and $p^* = p_0$ follows directly from Proposition 3. It remains to show that there is a unique equilibrium. Namely, we want to show that there exists a unique $b \in (0, 1)$ such that $(A^B = b, p^* = p_0)$ is an equilibrium strategy profile. First, define

$$W(a, b) \equiv p_0 R^b(a, 1) + (1 - p_0) R^b(a, 0)$$

where

$$R^b(a, \theta) \equiv \frac{1}{1 + \frac{1-R_0}{R_0} \frac{Pr(a, \theta | i=B, A^B=b)}{Pr(a, \theta | i=G, p^*=p_0)}}$$

is the unique reputation function that is consistent with the strategy profile $(A^B = b, p^* = p_0)$. I claim that there exists a unique $b \in (0, 1)$ such that $W(1, b) = W(\emptyset, b)$. First, note that

$$Pr(a, \theta | i = G, p^* = p_0) \in (0, 1) \text{ for all } a, \theta. \quad (29)$$

Now, I make two observations about W :

1. $W(1, b = 0) - W(\emptyset, b = 0) > 0$ and $W(1, b = 1) - W(\emptyset, b = 1) < 0$.

To show this, note that $Pr(1, \theta | i = B, A^B = 0) = 0$ for all θ . Thus, by (29), $R^{b=0}(1, \theta) = 1$ and $R^{b=0}(\emptyset, \theta) < 1$ for all θ . Thus, $W(1, b = 0) - W(\emptyset, b = 0) > 0$. One can analogously show that $W(1, b = 1) - W(\emptyset, b = 1) < 0$.

2. $W(1, b) - W(\emptyset, b)$ is continuous and strictly decreasing in b .

To show this, note that

$$R^b(1, 1) = \frac{1}{1 + \frac{1-R_0}{R_0} \frac{p_0 b}{1-F(1|\theta=1)}},$$

which is continuous and strictly decreasing in b . One can similarly show that $R^b(1, \theta) (R^b(\emptyset, \theta))$ is continuous and strictly decreasing (increasing) in b for all θ . The statement then follows from the definition of W .

1 and 2 imply that there exists a unique b such that $W(1, b) = W(\emptyset, b)$. It follows immediately that $(A^B = b, p^* = p_0)$ is the unique equilibrium strategy profile. $\square$

Proof of Corollary 4. Fix any $T < \infty$. Restrict attention to strategies where B and G never act after T . Furthermore, let $R(t, \theta) = 0$ for all $t > T, \theta$. For all $t \leq T$ let the strategies and R be those specified by an equilibrium under a deadline T , the existence of which is guaranteed by Proposition 1. It follows that this is a T -informative equilibrium. $\square$